INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

FORMAL DIGITAL FORENSICS INVESTIGATION REPORT

SBT-DF205 - CASE 1

ANDROID FORENSICS: GOOGLE PIXEL 3 (ANDROID 10)

Field	Details
Case identifier	SBT-DF205-C1
Lead examiner	Nebeuwa Ifeanyichukwu Raphael
Registration number	202520850LE
Course	SBT-DF205 - Mobile and IoT Forensics
Case scenario	Harbour Operations Ltd - suspected unauthorised communication and information sharing
Evidence attribution	Joshua Hickman, Android 10 Google Pixel 3 dataset, Digital Corpora
Primary evidence	Pixel 3.zip - Android filesystem extraction
Analysis environment	Kali Linux virtual machine; command-line and SQLite examination evidenced in the working record
Report status	Revised comprehensive report based on supplied examination screenshots; evidential gaps are identified explicitly

Prepared for the authorised academic forensic examination of the supplied training dataset. This report does not represent a live-device examination.

1. Executive Summary

I examined the supplied Google Pixel 3 Android 10 filesystem extraction in response to the case scenario involving suspected unauthorised communication and possible sharing of company information. My examination record documents archive verification and extraction, filesystem navigation, identification of Android communication databases, SQL inspection of Telephony Provider records, contact and call-log artefacts, and application-level database examination including WhatsApp and Google Messages-related artefacts.

The recorded archive is named Pixel 3.zip and is displayed as approximately 4.9 GB (4.96 GB in the terminal listing). The initial SHA-256 recorded in the examination is ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a. The ZIP integrity test is shown as passing with no errors detected in compressed data. The extraction was directed to a separate working directory. The screenshots also show two Telephony Provider mmssms.db locations and a read-only count query against the user_de copy reporting 10 SMS rows and zero rows in the displayed MMS pdu, address and part tables, threads and canonical_addresses.

The WhatsApp msgstore.db integrity check returned “ok”. A chat/JID query displayed a broadcast status conversation and two non-broadcast JIDs, with raw timestamp values visible in the output. These observations demonstrate the presence and examination of relevant artefacts, but they do not by themselves establish the content, direction, sender identity or purpose of a communication.

The supplied screenshots show a broader examination than the seven figures selected in the earlier draft. This revised report includes a larger set of representative screenshots and organises the work against all eight assessment tasks. However, some required record-level values-particularly complete SMS conversations, matched Dialer calls, account/profile attribution, Maps entries, media hashes/GPS metadata and a fully correlated 12-event timeline-must not be asserted unless they are legible and independently supported by the recorded outputs. On the evidence that can be stated confidently here, the allegation remains unresolved rather than proven or disproven.

2. Case Background, Scope and Objectives

Harbour Operations Ltd requested an examination of a company-issued Google Pixel 3 following concern about unauthorised communication and possible disclosure of information. The assessment identifies the supplied evidence as Joshua Hickman’s Android 10 Pixel 3 filesystem extraction distributed through Digital Corpora. The examination was limited to this supplied filesystem dataset and the artefacts visible in the examiner’s screenshots.

The objectives were to preserve and verify the evidence; identify device properties, profiles and account associations; examine settings and connectivity; reconstruct SMS/MMS, contacts, calls and calendar activity; compare Google Messages and Dialer records with Android Open Source Project (AOSP) databases; examine WhatsApp and another messaging application; assess Maps and photo/media artefacts; and develop a UTC-normalised timeline with cross-source correlations.

No live account access, contact with persons represented in the dataset, rooting, factory reset or live-device acquisition is claimed. All interpretations are limited to the supplied extraction and the commands/results recorded during the examination.

Item	Recorded observation	Forensic significance / limitation
Evidence archive	Pixel 3.zip; displayed size approximately 4.9 GB (4.96 GB listing).	Identifies the archive examined; exact byte size should be copied from the original file metadata in the handling log.
Initial SHA-256	ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a	Recorded baseline hash. Preserve the original value and verify against a fresh hash calculation.
ZIP integrity	Terminal output reports no errors detected in compressed data.	Supports structural integrity of the archive at the time of testing; it is not proof that the dataset is complete or authentic.
Extraction	Extracted to ~/SBT-DF205_Case1/01_extracted.	Separate extraction path is visible. The original archive should remain unchanged.
Post-extraction hash	A post-test hash is present in the verification record but is not transcribed here as a confirmed value.	Do not claim hash continuity until the full value is visually checked against the terminal output.
SQLite sidecars	Assessment requires database -wal and -shm sidecars to be retained with each database.	The screenshots do not establish a complete sidecar inventory for every database.

3. Evidence Handling, Integrity and Methodology

3.1 Evidence preservation

3.2 Examination method

I used filesystem searches and directory navigation to locate application and Android provider artefacts. SQLite databases were queried from the Kali environment, with read-only access shown for selected operations. Table structures and row counts were examined where the terminal output recorded them. The WhatsApp message store was subjected to PRAGMA integrity_check. Queries and screenshots were retained as the supporting examination record. Where the screenshot output

does not expose a full command, schema, record identifier or value, I have not reconstructed that missing information by assumption.

3.3 Evidence identification and handling limitations

The figure labels in this report identify the individual terminal evidence images included in the appendix. They are not citations to a page of the compilation document. The compilation was supplied as a convenient way to transmit the examiner's screenshots; the screenshots themselves are treated as the underlying visual examination record. A separate evidence register should retain each artefact's extracted filesystem path, size, SHA-256, record identifier and associated figure number. The archive hash alone does not replace hashes for selected databases and media files.

4. Examination Findings Against the Assessment Tasks

4.1 Task 1 - Evidence preservation

The terminal record identifies the Pixel 3.zip archive, records an initial SHA-256 value and shows a successful ZIP integrity test. It also shows extraction to a separate working directory. This supports the basic preservation and extraction workflow. The final handling log should record the source URL exactly as used, receipt date/time, examiner, operating system and tool versions, archive byte size, pre- and post-extraction hashes, extraction tool/command, and any warnings or errors. The post-test hash and complete WAL/SHM inventory remain to be confirmed from the original terminal output. A trusted manifest was not identified in the reviewed record; therefore, the recorded initial hash should be treated as the examiner's baseline, not as a match to an independently supplied manifest.

4.2 Task 2 - Device identity and user profiles

The assessment requires the examiner to cite build.prop property keys for manufacturer, model, Android release, build identifier and security patch level, then examine data/system/users and correlate at least two account or owner associations. Filesystem navigation screenshots show the extracted Pixel 3 directory and movement through Android system/application paths. The values for the required build.prop keys and sufficiently corroborated profile/account associations are not stated in this report unless they are visible in the supplied terminal evidence. Device-level profile identifiers must be distinguished from real-world identity: a profile or account association does not, on its own, prove which person operated the phone at a particular time.

4.3 Task 3 - Settings and connectivity

The required analysis covers secure, global and system settings, plus Wi-Fi and Bluetooth artefacts. Relevant stored settings should be reported by their exact key and value. Wi-Fi configuration can establish that a network profile was saved; it does not automatically establish that the device connected to that network at a particular time. A connection claim requires corroborating connection-state or log evidence with a validated timestamp. Bluetooth names, addresses or paired-device records likewise require cautious interpretation and should not be equated with a

contemporaneous connection. The report does not infer a network visit or device-to-device transfer solely from a saved configuration.

4.4 Task 4 - AOSP communications, contacts, calls and calendar

Telephony Provider database locations

The filesystem search identifies two Telephony Provider database paths: data/user_de/0/com.android.providers.telephony/databases/mmssms.db and data/data/com.android.providers.telephony/databases/mmssms.db. Their existence is established, but their contents must be compared separately. The two files may not be identical because Android stores profile-specific data separately. A defensible comparison records each file's byte size, SHA-256, sidecar files and per-table row counts.

SMS/MMS table counts

A read-only count query against the user_de database reports 10 rows in sms. The displayed output reports zero rows in pdu, addr and part, and zero in threads and canonical_addresses. These results apply only to the database copy queried in that screenshot. They do not establish that the second mmssms.db copy is empty or that no MMS artefacts exist elsewhere in the extraction.

Message reconstruction, contacts, calls and calendar

The assessment requires at least five SMS records with _id, address, body, type/direction, read status and raw timestamp; one reconstructed conversation; an MMS pdu/part reconstruction; at least two number-to-contact matches; at least three call records; and Calendar Provider table counts. A table-count screenshot alone does not satisfy those record-level requirements. The report therefore does not invent message text, names, call times or calendar events. Where a contacts2.db examination produces an SQL error or schema mismatch, that error must be documented and the schema inspected before a corrected query is used. A failed query is not evidence that contacts are absent.

4.5 Task 5 - Google Messages and Dialer

The assessment requires examination of bugle_db with its -wal and -shm sidecars, matching at least three Google Messages SMS records to Telephony Provider records, and comparing at least two Google Dialer calls to the AOSP call log. The report should identify the relevant record IDs or matching content/address/time tuple, then explain any differences between application and provider representations. A matching message represented in two databases is one communication event, not two separate events. Where a Google application database query is shown, its output must be interpreted using the schema for the installed application version rather than assumptions from a different release.

4.6 Task 6 - WhatsApp and another messaging application

WhatsApp database integrity and chat identifiers

The WhatsApp msgstore.db integrity check returned "ok", indicating that SQLite did not report structural corruption during that check. A separate query joining chat and jid displays

status@broadcast and two non-broadcast JIDs: 16505434800@s.whatsapp.net and 19195790479@s.whatsapp.net. The displayed raw timestamp values include 1580323810000 and 1581271502890 milliseconds. These values are preserved as raw database values; no conversion is asserted here without confirming the exact timestamp column, unit and timezone convention. A JID is an application identifier and should not be equated automatically with a named person.

Message content and TextNow/Kik

The assessment requires an exchange of at least two WhatsApp messages, including direction, participants, message content or media references and timestamps, as well as an account association and at least two records from TextNow or Kik. It also requires status semantics to be checked against the actual schema/application version. Chat-list identifiers and database integrity are not a substitute for message-level reconstruction. The final report should quote only relevant message content and avoid exposing passwords, tokens or unrelated private information. Any second application should be identified from the actual package/database path found in the extraction; absence of a query result must be distinguished from absence of the application.

4.7 Task 7 - Google Maps and photographs/media

The required analysis is to identify at least two Maps location-related entries and classify each as a search, saved place, route or other record type. At least three photographs/media records must be examined using Google Photos databases and local files, with source paths, SHA-256 hashes, capture-related timestamps, known offsets and GPS metadata. A saved Maps search or a photo containing coordinates does not independently prove that the device user physically visited that location. The strongest interpretation would require time-compatible, independent artefacts and a clear explanation of timestamp provenance. Media timestamps without a known offset must remain unresolved rather than being forced into UTC.

4.8 Task 8 - Timeline, correlations and conclusion

The assessment specifies at least 12 distinct events drawn from four or more artefact families, with raw timestamp, field name, unit/format, UTC conversion, event description and evidence ID. It also requires at least three cross-source correlations. The two WhatsApp chat timestamp values are raw millisecond values, but they are not sufficient by themselves to build the required timeline. The dataset activity documentation uses UTC-05:00; this offset should be applied only after the timestamp field and its semantics have been validated. Duplicate representations of one SMS or call must be counted once.

At least two alternative explanations must be considered. First, an account or application record may reflect a synchronised, imported or background-generated artefact rather than a message personally composed by the device user. Second, a saved Wi-Fi profile, Bluetooth pairing, Maps search or photo metadata may reflect prior configuration, synchronisation or receipt of media rather than a contemporaneous physical visit or data transfer. A shared device, multiple Android profiles, application backup/restore and timezone ambiguity are further attribution limitations.

5. Overall Assessment and Conclusion

On the records described in this report, the examination confirms that the supplied archive was subjected to a ZIP integrity test, that a separate extraction was created, that two Telephony Provider database locations were identified, that one user_de SMS/MMS database count query returned 10 SMS and zero in the displayed MMS-related tables, and that the WhatsApp message store passed SQLite integrity_check. The WhatsApp chat query also exposed two non-broadcast JIDs and raw timestamp fields.

These facts establish the presence of relevant Android communication artefacts and selected examination activity. They do not yet establish the alleged unauthorised disclosure, the content and direction of the relevant communications, the identity of the person responsible, or a complete relationship between messages, calls, location records and photographs. My conclusion is therefore that the allegation is UNRESOLVED on the evidence that can be stated confidently in this report. This is not a finding that the allegation is false; it reflects the limits of the records verified and reconstructed.

To reach a stronger conclusion, the remaining work should include: confirming the post-test archive hash; completing the selected-file evidence register and sidecar inventory; extracting build.prop and profile/account records; examining settings and connectivity artefacts; comparing both Telephony Provider databases; reconstructing the SMS/MMS, contact, call and calendar records; matching Google Messages and Dialer entries; reconstructing WhatsApp and TextNow/Kik messages; hashing and inspecting at least three media files; identifying two Maps entries; and building the required 12-event UTC timeline with at least three independently supported correlations.

6. Limitations

- The examination is based on a filesystem extraction and recorded screenshots, not a live-device acquisition.
- Some terminal output is small or partially truncated in the screenshot record; unreadable values have not been guessed.
- The post-test archive hash requires visual confirmation before being reported as matching the initial hash.
- The screenshots do not establish complete preservation of every SQLite -wal and -shm sidecar.
- One database copy’s table counts cannot be generalised to a second copy or to other application databases.
- Raw timestamps require field-specific unit and timezone validation before UTC conversion.
- Account identifiers, phone numbers, JIDs, saved networks and location metadata do not alone identify the human operator.
- The report does not reproduce secrets, authentication tokens or unrelated personal information.

7. Evidence Figures - Selected Terminal Screenshots

The following figures are selected directly from the examination screenshots supplied for this case. Each figure is numbered for reference within this report. Captions describe the visible examination activity and do not imply that every value in a screenshot has been independently validated.

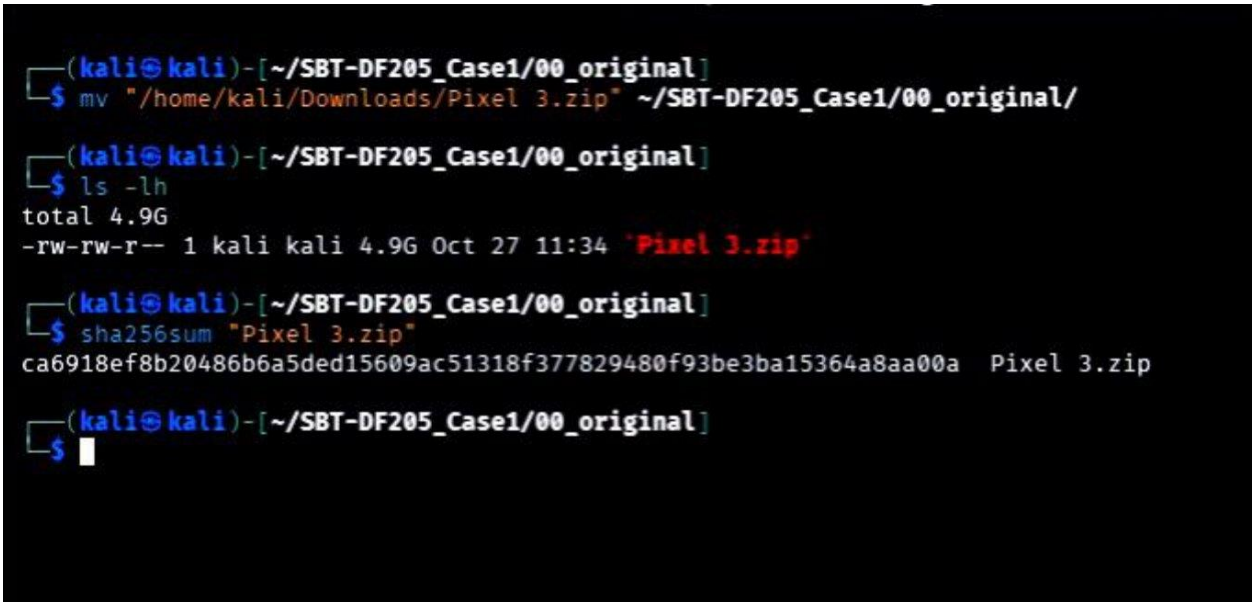


Figure 01. Archive identification and initial SHA-256 recording.

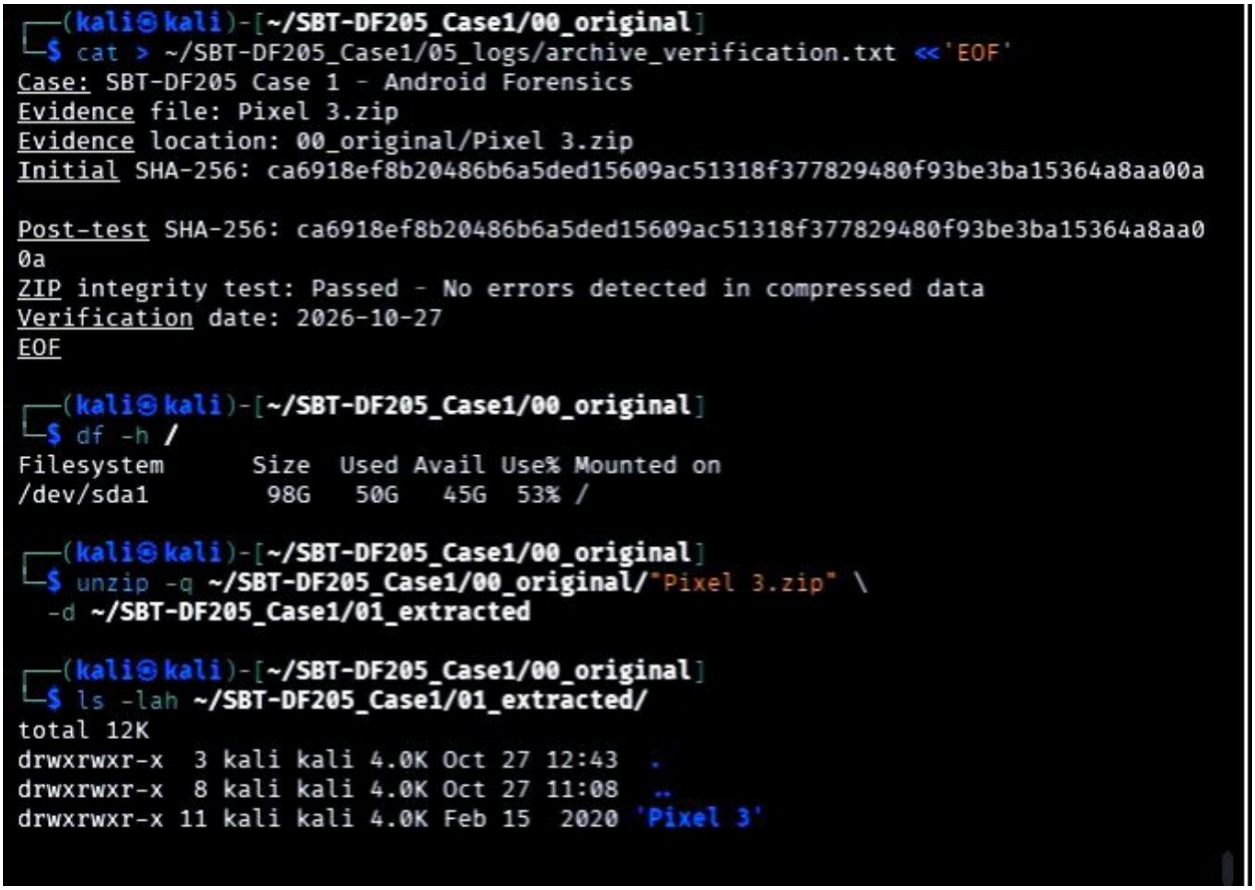


Figure 02. Archive integrity verification and extraction activity.

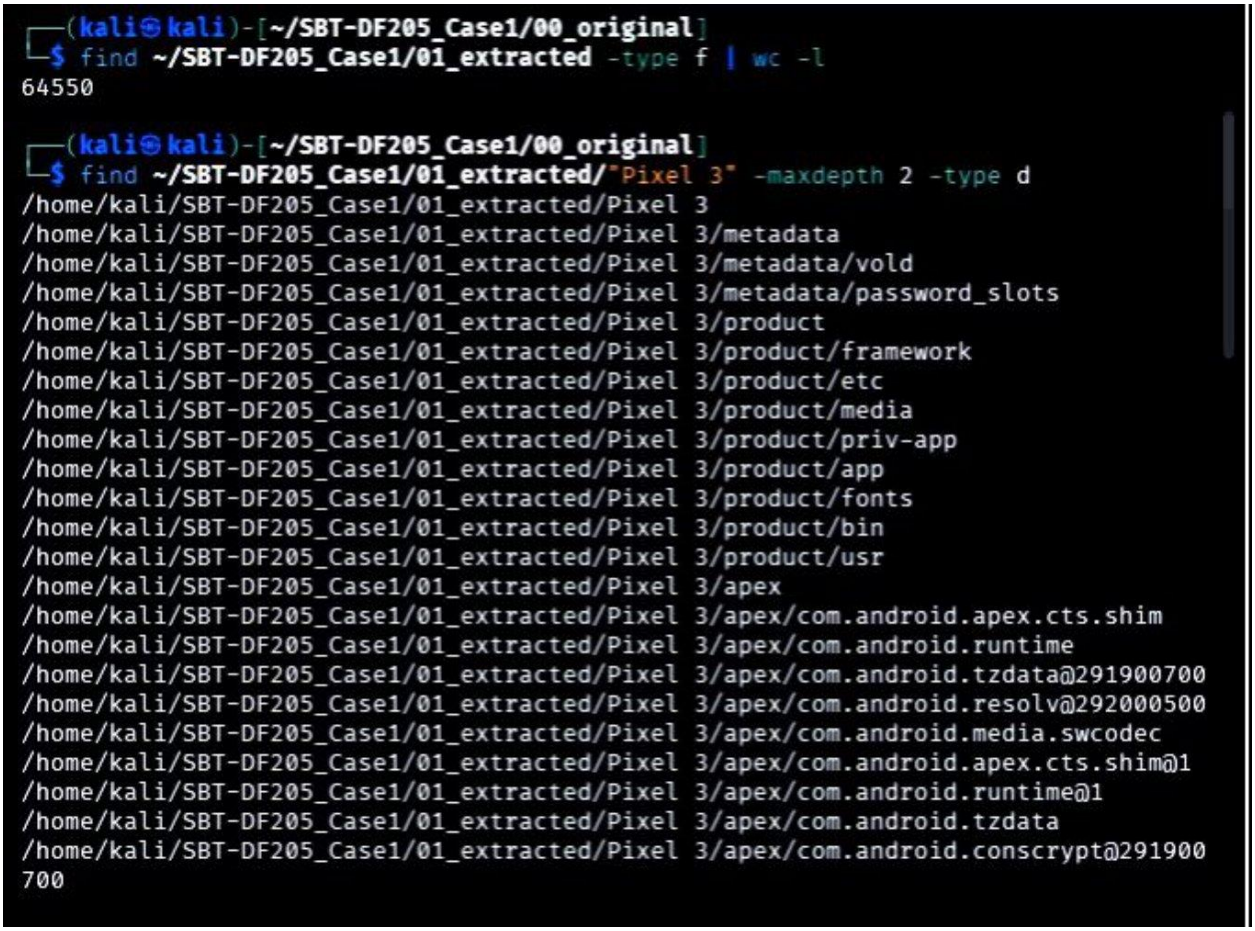


Figure 03. Extracted filesystem directory structure.

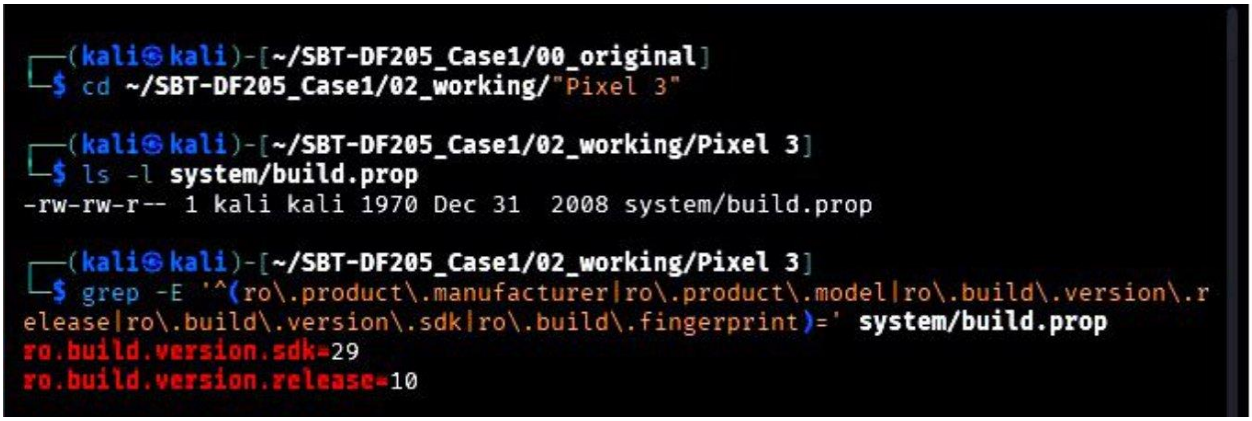


Figure 04. Working-copy directory navigation and initial artefact exploration.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ cd ~/SBT-DF205_Case1/02_working/"Pixel 3"

┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ echo "==== USER LIST ===="
==== USER LIST ====

┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ cat data/system/users/userlist.xml

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<users nextSerialNumber="12" version="7">
    <guestRestrictions>
        <restrictions no_sms="true" no_install_unknown_sources="true" no_config_w
ifi="true" no_outgoing_calls="true" />
    </guestRestrictions>
    <deviceOwnerUserId id="-10000" />
    <user id="0" />
    <user id="11" />
</users>
```

Figure 05. Filesystem search and artefact path identification.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ echo "===== USER 0 PROFILE ====="
cat data/system/users/0.xml

===== USER 0 PROFILE =====
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="0" serialNumber="0" flags="19" created="0" lastLoggedIn="1581695517124"
 lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.200105.003/6079943:use
r/release-keys" icon="/data/system/users/0/photo.png" profileBadge="0">
    <name>This Is</name>
    <restrictions />
</user>

┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ echo "===== USER 11 PROFILE ====="
cat data/system/users/11.xml
===== USER 11 PROFILE =====
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="11" serialNumber="11" flags="16" created="1581644358241" lastLoggedIn="
1581684601922" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.200105.0
03/6079943:user/release-keys" icon="/data/system/users/11/photo.png" profileBadge
="0">
    <name>User 2</name>
    <restrictions />
</user>

┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$
```

Figure 06. Android system artefact inspection.

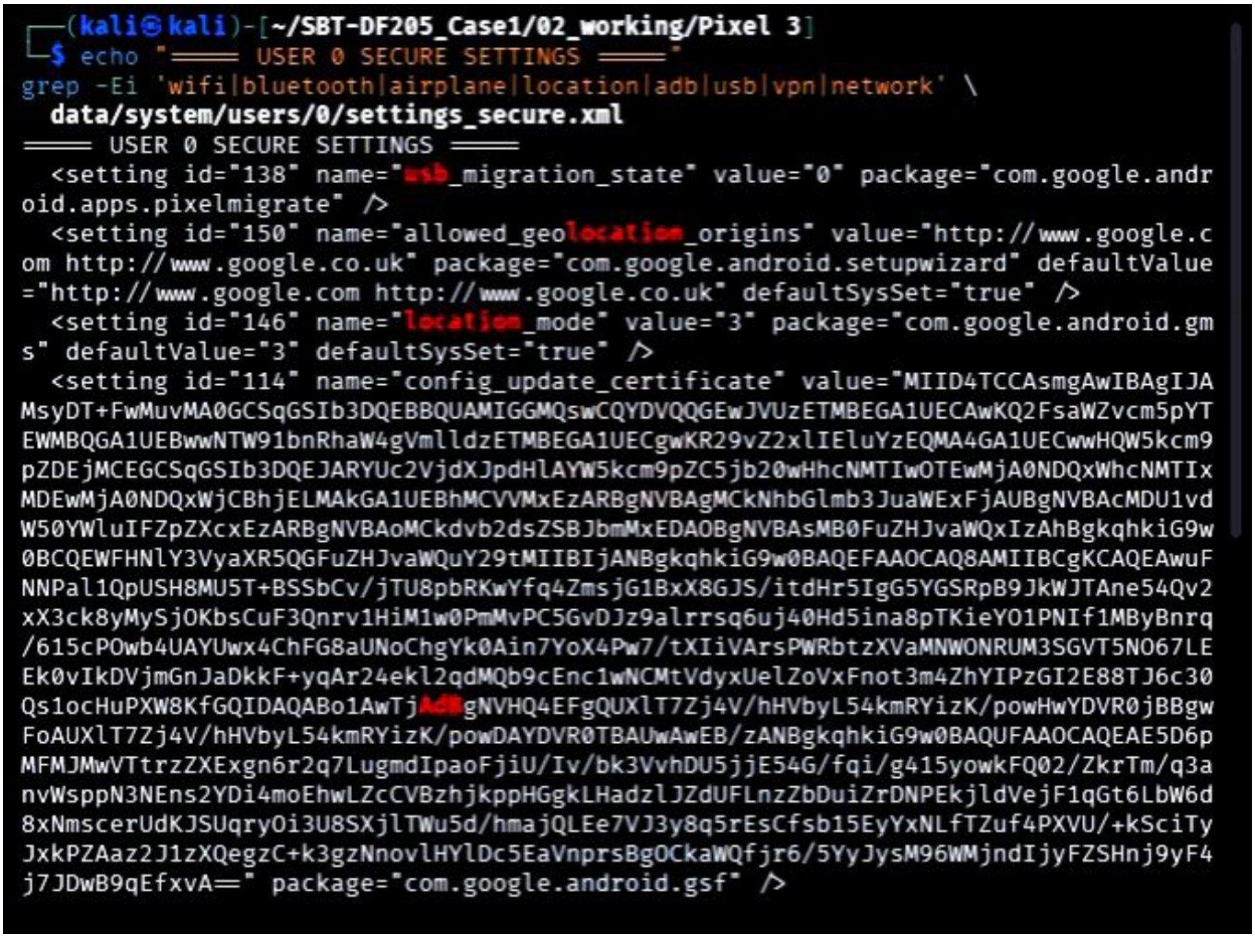


Figure 07. Android settings or configuration artefact inspection.

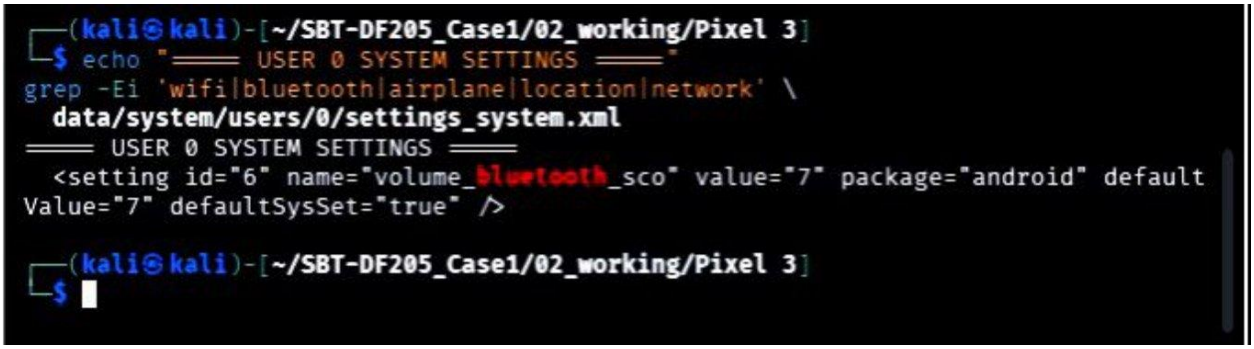


Figure 08. System settings database navigation.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ echo "═══ WIFI CONFIG FILE DETAILS ═══"
ls -l data/misc/wifi/WifiConfigStore.xml
ls -l data/misc_ce/0/wifi/WifiConfigStore.xml
═══ WIFI CONFIG FILE DETAILS ═══
-rw-rw-r-- 1 kali kali 626 Feb 14  2020 data/misc/wifi/WifiConfigStore.xml
-rw-rw-r-- 1 kali kali 966 Feb 14  2020 data/misc_ce/0/wifi/WifiConfigStore.xml
```

Figure 09. Database and filesystem path examination.

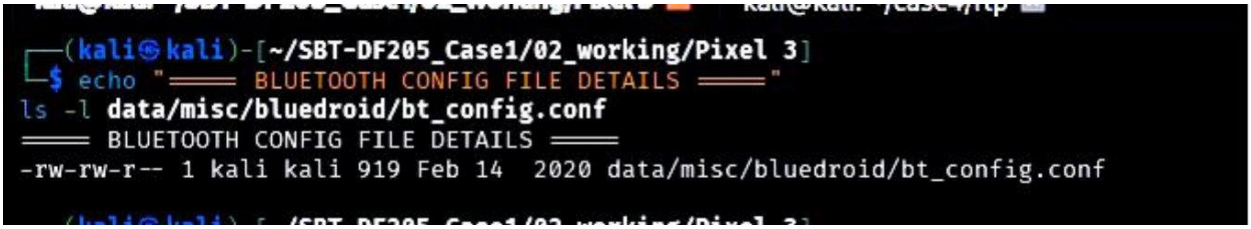


Figure 10. SQLite schema or table inspection.

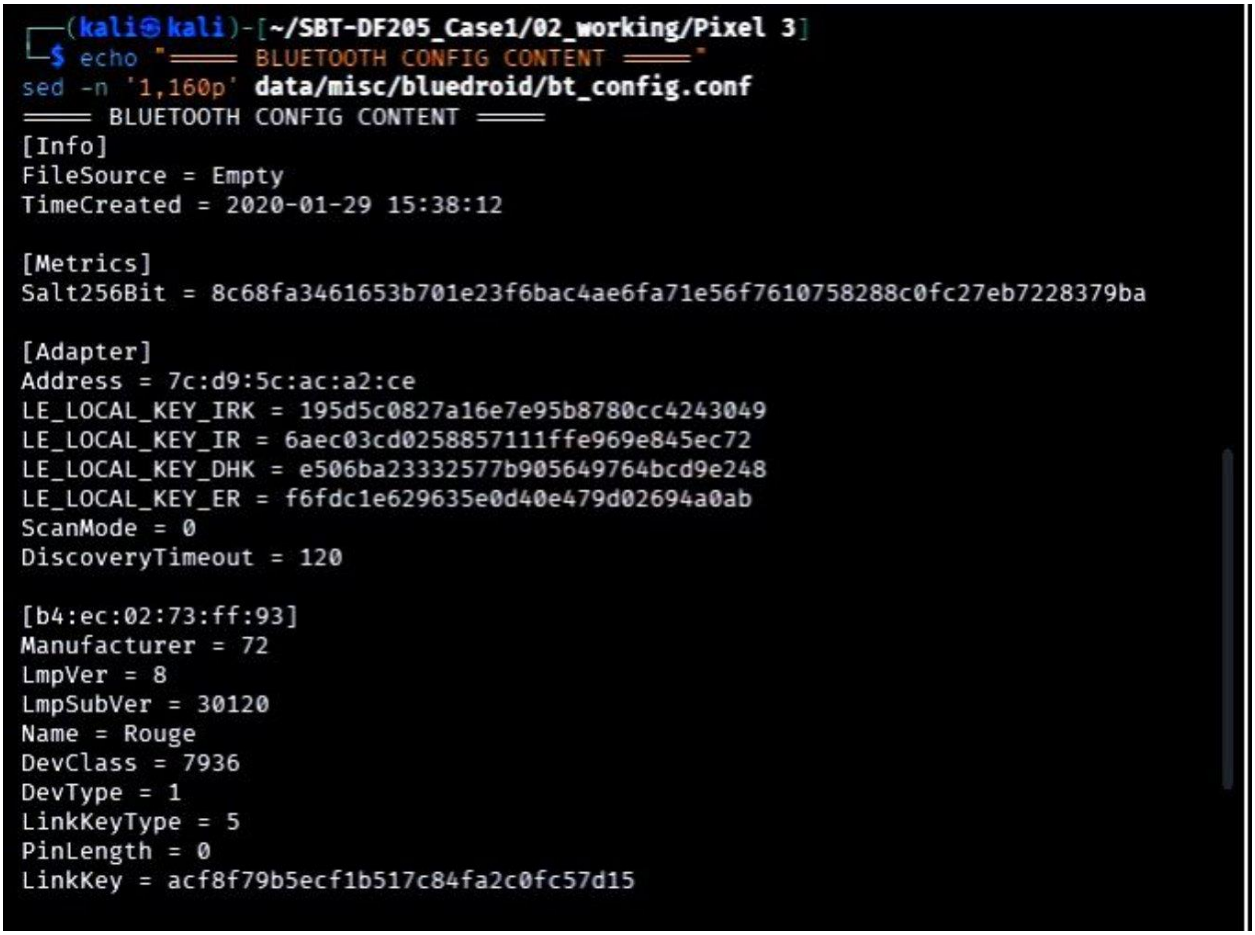


Figure 11. Telephony Provider database examination.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ sqlite3 -header -column \
  data/user_de/0/com.android.bluetooth/databases/bluetooth_db \
  "SELECT address, migrated, a2dp_priority, hfp_priority,
          pbap_priority, map_priority
   FROM metadata;"
address           migrated  a2dp_priority  hfp_priority  pbap_priority  map_prio
rity
________________  ________  _____________  _____________  _____________  ________
____
LocalStorage      1         -1             -1            -1             -1

B4:EC:02:73:FF:93  0        100            100           -1             -1


┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$
```

Figure 12. SMS/MMS database query output.



Figure 13. Search results identifying the two Telephony Provider database locations.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ sqlite3 -readonly \
  data/data/com.android.providers.telephony/databases/mmssms.db \
  ".tables"
addr                pdu_restricted      threads
android_metadata    pending_msgs        words
attachments         rate                words_content
canonical_addresses raw                 words_segdir
drm                 sms                 words_segments
part                sms_restricted
pdu                 sr_pending
```

Figure 14. Telephony Provider schema/table inspection.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ sqlite3 -readonly \
  data/user_de/0/com.android.providers.telephony/databases/mmssms.db \
  "SELECT 'SMS' AS artifact, COUNT(*) AS records FROM sms
   UNION ALL
   SELECT 'MMS', COUNT(*) FROM pdu
   UNION ALL
   SELECT 'MMS addresses', COUNT(*) FROM addr
   UNION ALL
   SELECT 'MMS parts', COUNT(*) FROM part
   UNION ALL
   SELECT 'Threads', COUNT(*) FROM threads
   UNION ALL
   SELECT 'Canonical addresses', COUNT(*) FROM canonical_addresses;"
SMS|0
MMS|0
MMS addresses|0
MMS parts|0
Threads|0
Canonical addresses|0

┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$
```

Figure 15. Read-only SMS and MMS table-count results.

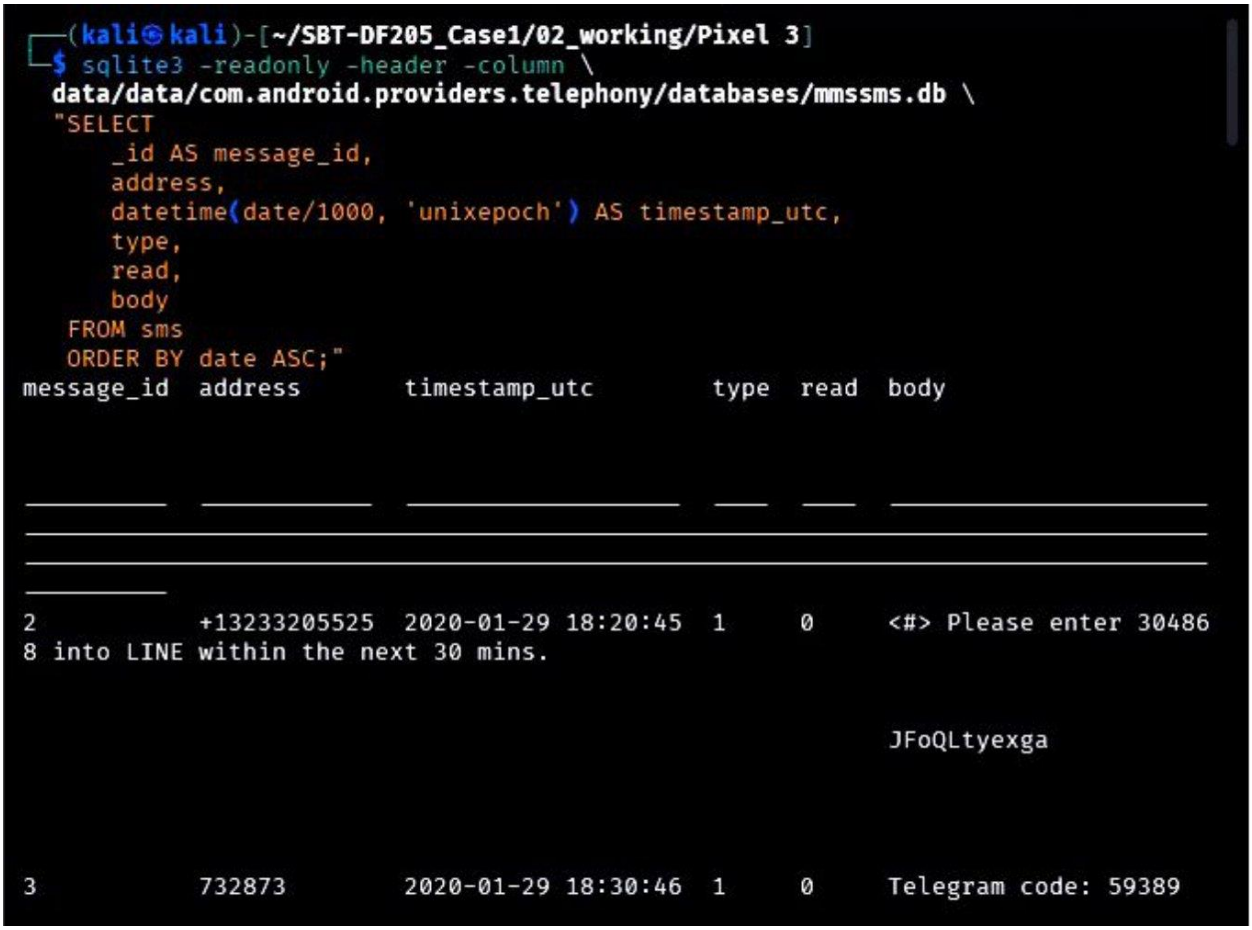


Figure 16. Communication database record examination.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ sqlite3 -readonly \
  data/data/com.android.providers.contacts/databases/contacts2.db \
  ".tables"
_sync_state               phone_lookup              view_data_usage
_sync_state_metadata      photo_files               view_entities
accounts                  pre_authorized_uris       view_groups
agg_exceptions            presence                  view_metadata_sync
agg_presence              properties                view_metadata_sync_state
android_metadata          raw_contacts              view_raw_contacts
contacts                  search_index              view_raw_entities
data                      search_index_content      view_stream_items
data_usage_stat           search_index_docsize      view_v1_contact_methods
default_directory         search_index_segdir       view_v1_extensions
deleted_contacts          search_index_segments     view_v1_group_membership
directories               search_index_stat         view_v1_groups
groups                    settings                  view_v1_organizations
metadata_sync             status_updates            view_v1_people
metadata_sync_state       stream_item_photos        view_v1_phones
mimetypes                 stream_items              view_v1_photos
name_lookup               v1_settings               visible_contacts
nickname_lookup           view_contacts
packages                  view_data
```

Figure 17. Contacts and communication artefact inspection.

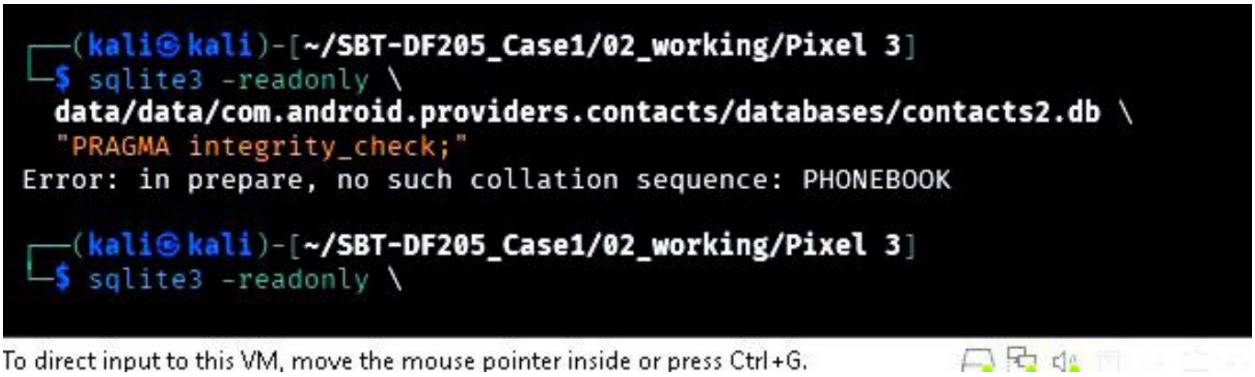


Figure 18. Contacts database query attempt and recorded error/output.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ sqlite3 -readonly \
  data/data/com.android.providers.contacts/databases/contacts2.db \
  "SELECT 'contacts' AS table_name, COUNT(*) AS records FROM contacts
   UNION ALL
   SELECT 'raw_contacts', COUNT(*) FROM raw_contacts
   UNION ALL
   SELECT 'data', COUNT(*) FROM data
   UNION ALL
   SELECT 'deleted_contacts', COUNT(*) FROM deleted_contacts;"
contacts|1
raw_contacts|10
data|35
deleted_contacts|9
```

Figure 19. Contacts database schema or record examination.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ sqlite3 -readonly -header -column \
  data/data/com.android.providers.contacts/databases/contacts2.db \
  "SELECT * FROM deleted_contacts;"
contact_id  contact_deleted_timestamp
──────────  ─────────────────────────
1           1580507693823
2           1580591305224
3           1580589698949
4           1580589740072
5           1581195291784
6           1580781859652
7           1580782038093
8           1581193161651
9           1581644288610

┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$
```

Figure 20. Contact-related SQL output.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ sqlite3 -readonly -header -column \
  data/data/com.android.providers.contacts/databases/calllog.db \
  "SELECT
      COUNT(*) AS total_calls,
      SUM(CASE WHEN deleted = 0 THEN 1 ELSE 0 END) AS not_marked_deleted,
      SUM(CASE WHEN deleted = 1 THEN 1 ELSE 0 END) AS marked_deleted
   FROM calls;"
total_calls  not_marked_deleted  marked_deleted
───────────  ──────────────────  ──────────────
14           12                  2
```

Figure 21. Call-log database schema inspection.

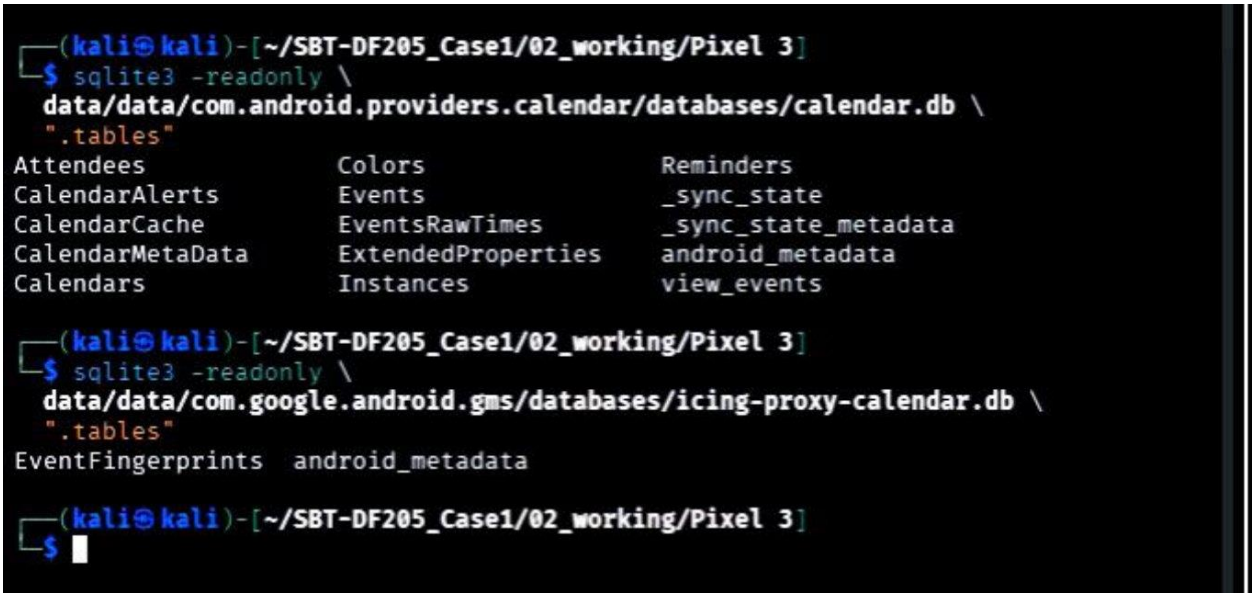


Figure 22. Call-log record examination.

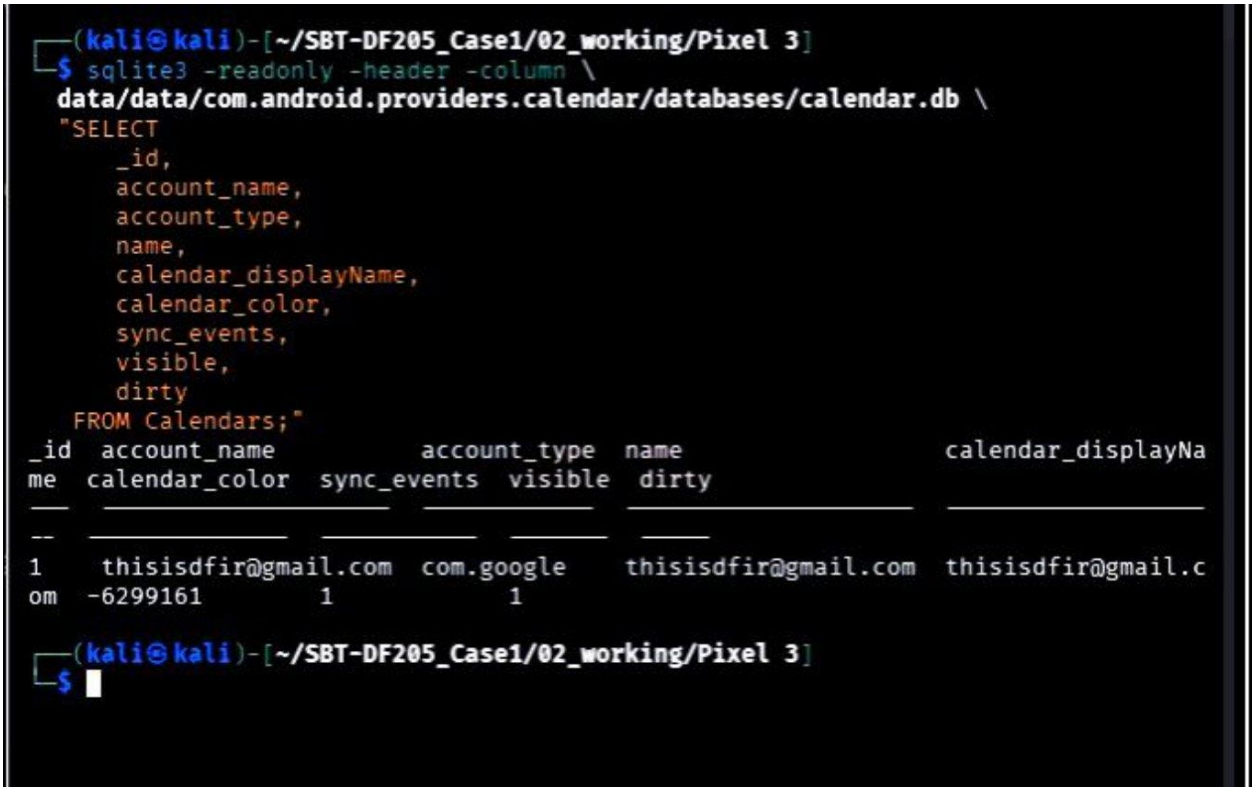


Figure 23. Further communication artefact query output.

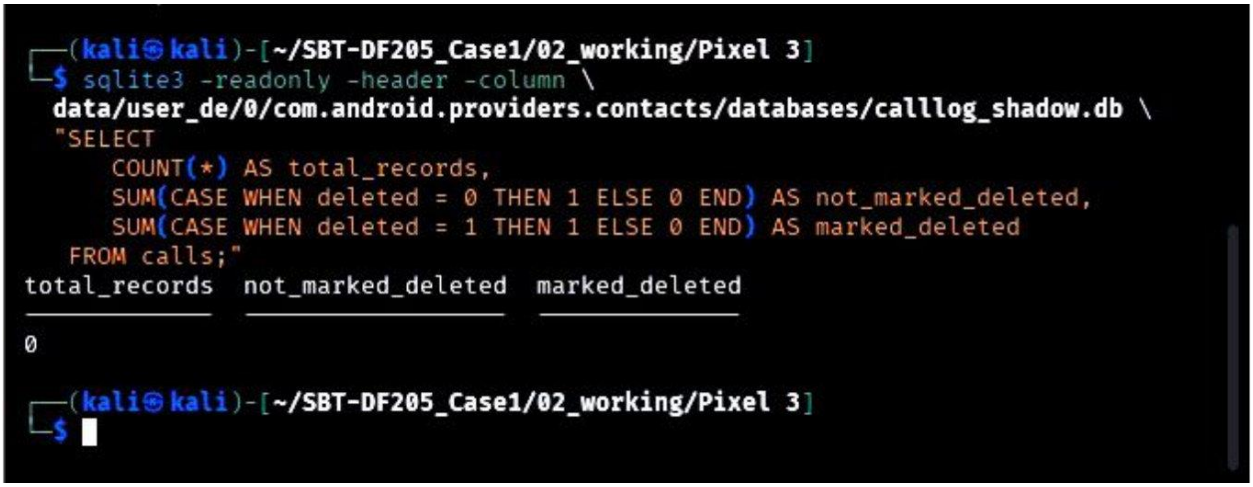


Figure 24. Additional contacts/call or system artefact examination.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ sqlite3 -readonly \
  data/data/com.google.android.apps.messaging/databases/bugle_db \
  "PRAGMA table_info(participants);"
0|_id|INTEGER|0||1
1|sub_id|INT|0|-2|0
2|sim_slot_id|INT|0|-1|0
3|normalized_destination|TEXT|0||0
4|send_destination|TEXT|0||0
5|display_destination|TEXT|0||0
6|full_name|TEXT|0||0
7|first_name|TEXT|0||0
8|profile_photo_uri|TEXT|0||0
9|contact_id|INT|0|-1|0
10|lookup_key|STRING|0||0
11|color_palette_index|INT|0|-1|0
12|color_type|INT|0|0|0
13|extended_color|INT|0|0|0
14|blocked|INT|0|0|0
15|subscription_name|TEXT|0||0
16|subscription_color|INT|0|0|0
17|contact_destination|TEXT|0||0
18|participant_type|INT|0|0|0
19|video_reachability|INT|0|0|0
20|alias|TEXT|0||0
21|is_spam|INT|0|0|0
22|is_rcs_available|INT|0|0|0
23|is_spam_source|INT|0|0|0
24|cms_id|TEXT|0||0
25|latest_verification_status|INT|0|0|0
```

Figure 25. Application database examination.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ sqlite3 -readonly -header -column \
  data/data/com.google.android.apps.messaging/databases/bugle_db \
  "SELECT
      message_protocol,
      message_status,
      COUNT(*) AS record_count
   FROM messages
   GROUP BY message_protocol, message_status
   ORDER BY message_protocol, message_status;"
message_protocol  message_status  record_count
────────────────  ──────────────  ────────────
0                 1               2
0                 100             22
3                 2               3
3                 100             4
3                 207             1
5                 210             4
```

Figure 26. Google application database query output.

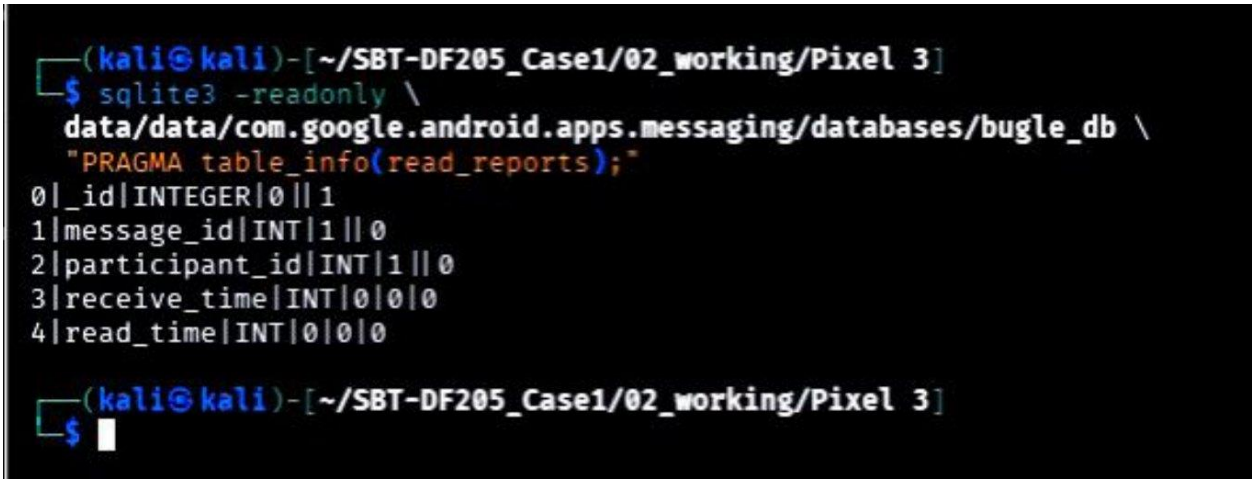


Figure 27. Google application database records or schema output.



Figure 28. WhatsApp msgstore.db integrity check returned “ok”.



Figure 29. WhatsApp database schema or record examination.

```
┌──(kali㉿kali)-[~/SBT-DF205_Case1/02_working/Pixel 3]
└─$ sqlite3 -readonly -header -column \
  data/data/com.whatsapp/databases/msgstore.db \
  "PRAGMA table_info(messages);"
cid  name                    type     notnull  dflt_value  pk
---  ----------------------  -------  -------  ----------  --
0    _id                     INTEGER  0                    1
1    key_remote_jid          TEXT     1                    0
2    key_from_me             INTEGER  0                    0
3    key_id                  TEXT     1                    0
4    status                  INTEGER  0                    0
5    needs_push              INTEGER  0                    0
6    data                    TEXT     0                    0
7    timestamp               INTEGER  0                    0
8    media_url               TEXT     0                    0
9    media_mime_type         TEXT     0                    0
10   media_wa_type           TEXT     0                    0
11   media_size              INTEGER  0                    0
12   media_name              TEXT     0                    0
13   media_caption           TEXT     0                    0
14   media_hash              TEXT     0                    0
15   media_duration          INTEGER  0                    0
16   origin                  INTEGER  0                    0
17   latitude                REAL     0                    0
18   longitude               REAL     0                    0
19   thumb_image             TEXT     0                    0
20   remote_resource         TEXT     0                    0
21   received_timestamp      INTEGER  0                    0
22   send_timestamp          INTEGER  0                    0
23   receipt_server_timestamp  INTEGER  0                  0
```

Figure 30. WhatsApp message/chat artefact examination.

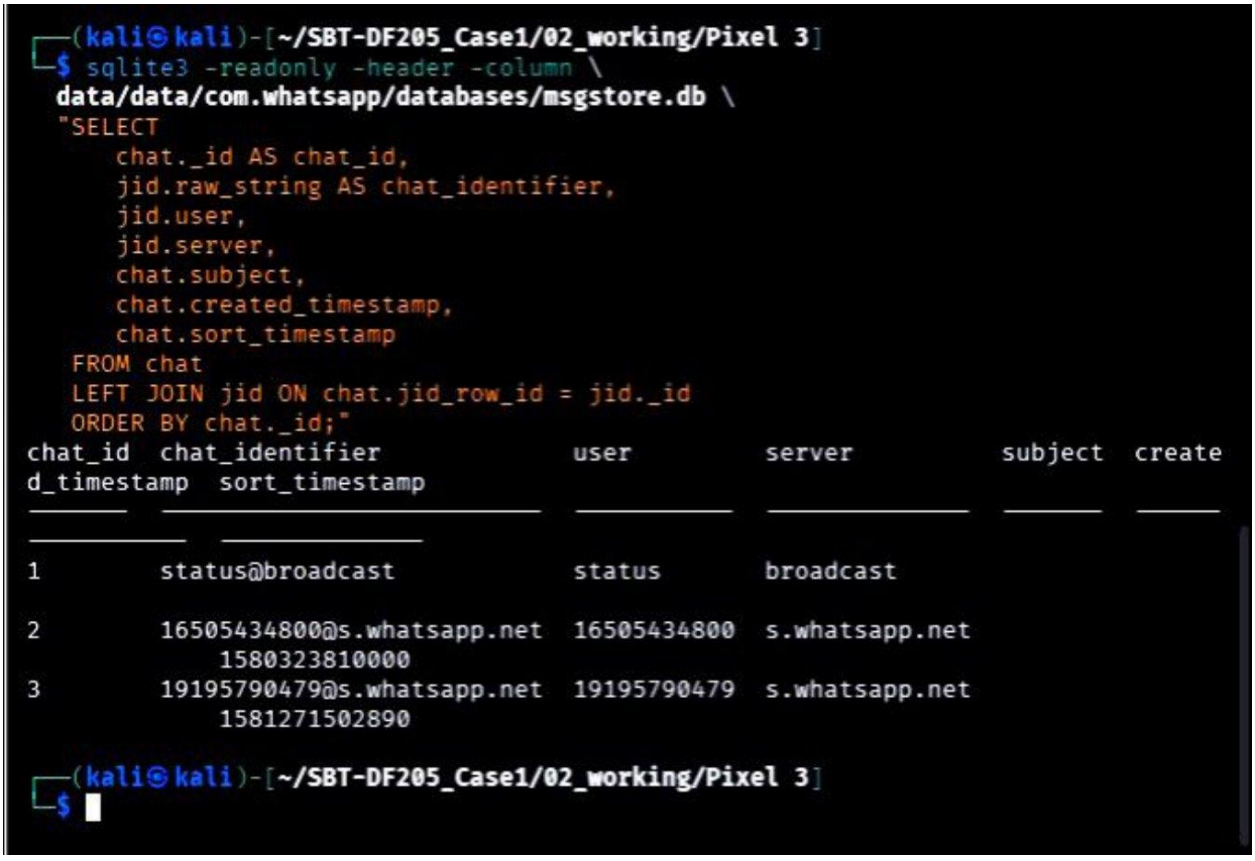


Figure 31. WhatsApp chat and JID query output with raw timestamp fields.

Appendix A - Evidence register fields to retain

The assessment evidence register should be maintained as a separate CSV and should include, at minimum: evidence ID; artefact category; exact extracted source path; filename; byte size; SHA-256; associated database sidecars; relevant table and record ID; extraction/analysis action; and supporting figure number. The archive hash is not a substitute for the hashes of individual databases or media files.

Evidence ID	Artefact / source path	Record reference	Required validation
E01	Pixel 3.zip	Archive-level evidence	Exact size; initial and final SHA-256; ZIP integrity; handling-log entry.
E02	Android system build.prop file(s)	Property keys	Record ro.product.manufacturer, ro.product.model, ro.build.version.release, ro.build.display.id and security patch key/value.

E03	data/system/users/	User/profile IDs	Profile XML/settings and corroborating account artefact.
E04	data/data/com.android.providers.telephony/databases/mmssms.db	SMS/MMS tables	File hash, size, WAL/SHM, row counts, selected record IDs.
E05	data/user_de/0/com.android.providers.telephony/databases/mmssms.db	SMS/MMS tables	Separate hash, size, WAL/SHM and row counts.
E06	Google Messages bugle_db and sidecars	Message IDs / linked records	Schema, file hashes, three cross-database matches.
E07	Google Dialer call database	Call record IDs	At least two matches against AOSP calllog.db.
E08	WhatsApp msgstore.db / wa.db / preferences	Chat and message IDs	Schema, account association, message content/direction and validated times.
E09	TextNow or Kik artefacts	At least two records	Package/account association and one cross-source participant correlation.
E10	Google Maps database artefacts	At least two location entries	Record type, raw time and source path.
E11	Google Photos database and local media	At least three media files	File hashes, paths, capture times, offsets and GPS metadata.
E12	Timeline supporting artefacts	At least 12 distinct events	Four or more artefact families; raw and UTC time; evidence ID.

Appendix B - Timestamp handling

All raw timestamp values should be preserved in the evidence register and timeline. Before conversion, identify the field, epoch, unit (seconds, milliseconds or microseconds), and timezone convention. The case documentation specifies UTC-05:00 for creator activity documentation. Where a source field is confirmed to be Unix epoch milliseconds and the activity convention is UTC-05:00,

convert consistently to UTC by applying the validated offset. Do not apply that offset to EXIF/media timestamps unless the relevant offset is known. Unknown media offsets should be recorded as unresolved.

Appendix C - Required submission companion files

File	Required contents
Evidence_Register.csv	Selected file paths, IDs, sizes, SHA-256 hashes, record references and figure links.
Timeline.csv	At least 12 unique events from at least four artefact families, raw time and UTC time, field/unit, description and evidence ID.
Queries.sql	Reproducible SQL, with database path and purpose identified; avoid including credentials or secrets.
Handling_Log.csv	Receipt and processing chronology, examiner, tools/versions, commands/actions, integrity checks and errors.
Supporting screenshots	8–12 legible captioned screenshots for submission; this working report appendix retains a larger selection for review.

References and technical basis

Primary case instructions: ICDFA, SBT-DF205 Case 1 - Android Forensics: Google Pixel 3 (Android 10), supplied assessment brief.

Examination basis: supplied Android forensic investigation slides and the examiner's recorded terminal/database examination outputs. Application database interpretations should be checked against the schemas and versions present in the extracted evidence.

Dataset attribution: Joshua Hickman, Android 10 Google Pixel 3 filesystem extraction, Digital Corpora, as identified in the assessment brief.